

MUHAMMAD ALI

AI/ML Engineer & Full Stack Developer | Speech & NLP Researcher

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EXECUTIVE SUMMARY

AI/ML Engineer and Full-Stack Developer specializing in state-of-the-art automatic speech recognition (ASR), computer vision, and high-performance neural network pipelines. Lead researcher of BaltiVoice (arXiv:2606.03504), an open-access speech corpus and fine-tuned OpenAI Whisper ASR baseline for low-resource languages. Experienced in building end-to-end scalable web platforms using React, TypeScript, Node.js, and PyTorch.

FEATURED RESEARCH & PUBLICATIONS

BaltiVoice: Speech Corpus & Fine-Tuned Whisper ASR System (2025)

Lead Author | Published on arXiv (arXiv:2606.03504) & HuggingFace Hub

½ Constructed 16.8-hour validated audio corpus (10,060 clips across 136 speakers) for the low-resource Balti language (bft).

½ Fine-tuned OpenAI Whisper-small model, reducing Word Error Rate (WER) from 159.19% to 26.74% (-83% relative reduction) and Character Error Rate (CER) to 8.67%.

½ Released open-source dataset, PyTorch training code, and live Gradio demo on HuggingFace Spaces.

TECHNICAL SKILLS

AI / ML & ASR:	PyTorch, OpenAI Whisper, HuggingFace Transformers, TorchAudio, OpenCV, Librosa
Frontend & UI:	React, TypeScript, Next.js, Tailwind CSS, Motion / Framer Motion, Vite
Backend & Cloud:	Node.js, Express, Python, REST APIs, Docker, Git / GitHub, MLOps
Databases & Tools:	PostgreSQL, MongoDB, Firebase, Gradio, Linux, Jupyter Notebooks

PROJECTS & ENGINEERING EXPERIENCE

BaltiVoice ASR Speech Recognition Engine

Lead ML Researcher & Developer

Engineered automated data preprocessing, audio spectrogram filterbanks (16kHz WAV), speaker-disjoint GroupShuffleSplit partitioning, and Whisper-small transfer learning pipeline.

NeuroVision AI Platform & Custom Neural Architectures

Full-Stack AI Developer

Built scalable web interfaces and backend proxy services integrating custom computer vision models with real-time analytics dashboards.

Design Systems & High-Performance Web Applications

Frontend Engineer

Designed modern dark-aesthetic responsive web interfaces with sub-second page loads, custom micro-interactions, and responsive layout architectures.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Software Engineering (BSSE)

Specialization in Machine Learning, Speech Processing, & Web Architecture

Certifications: Deep Learning Specialization ½ PyTorch for Audio & NLP ½ Full-Stack Web Development